

ETC5512: Regular Expressions and Tidy Data

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Learning Objectives

In this tutorial, you will be using the Australia Census data and be learning about:

- Structuring your projects
- Wrangling messy data into tidy data
- Practicing your string manipulation

Note: You should have completed all the [startR modules](#). We will assume you are up to date with the content in these moving forward.

Before your tutorial

For today’s class:

1. Set up a Quarto and project folder

Reminder of the details

We expect you to use an R projects and Quarto from now on.
Below is an example of how to get started.

- RStudio > New Project > New Directory > New Project

Save it in a directory of your choosing.

You can call the project what you like, but we recommend `exercise-04`.

- Create folders called `data` and `analysis` (or similarly named) in the project root directory.
- Create an `exercise-04.qmd` under the `analysis` (or similarly named) folder.
- Save your analysis from the exercises into the file, `exercise-04.qmd`.

2. Getting the census data

Instructions to get the data

- Go to <https://www.abs.gov.au/census/find-census-data/datapacks> selecting the following options:
- 2021 Census Datapacks - General Community Profile - All geographies - Vic

The downloaded file would be called `2021_GCP_ALL_for_Vic_short-header.zip`.

- Unzip this file and save the folder in the `data` folder you set up.

3. Installing the `here` R-package

Why use `here`?

While R Projects do most of the heavy lifting when it comes to specifying a working directory, `here` provides a more robust coding solution for referencing files.

```
install.packages("here")
```

4. Check your project structure

Why check the project structure?

Common errors occur from not referencing files correctly, particularly when reading data or rendering Quarto documents.

Your project structure should look something like below.

```
tutorial-04
  analysis
```

```

exercise-04.Rmd
data
  2021_GCP_ALL_for_Vic_short-header
    2021\ Census\ GCP\ All\ Geographies\ for\ VIC
      SA1
        VIC
          2021Census_G17A_VIC_SA1.csv
          2021Census_G17B_VIC_SA1.csv
          2021Census_G17C_VIC_SA1.csv
        STE
          VIC
            2021Census_G17A_VIC_STE.csv
            2021Census_G17B_VIC_STE.csv
            2021Census_G17C_VIC_STE.csv
tutorial-04.Rproj

```

- Note that it's not desirable to have spaces in your file or folder name so we use underscores.

Exercise 1

Practice using regular expressions

Consider the following vector A and B.

```

A <- c("F_300_399_55_64_yrs",
      "F_1_149_35_44_yrs",
      "F_150_299_25_34_yrs",
      "M_800_999_75_84_yrs",
      "M_400_499_65_74_yrs")

B <- c(A,
      "F_1_149_15_19_yrs",
      "F_150_299_85ov",
      "M_1500_1749_85ov",
      "M_Neg_Nil_income_20_24_yrs",
      "M_150_299_25_34_yrs")

```

The first letter has information on the sex, followed by minimum income, maximum income, minimum age, and maximum age separated by _, with some exceptions for certain age and income categories (e.g. Neg_Nil is negative or nil income, 85ov is 85 years or over).

We want to wrangle this vector to extract the variables and later make them their own columns in a data frame. Try using the functions `str_split()`, `str_replace()` and or `str_remove()` to do this.

Hint

```
# some examples to help you
library(tidyverse)
str_split("A_B_C_D", "_")
```

```
[[1]]
[1] "A" "B" "C" "D"
```

```
str_replace("Neg_nil_income", "Neg", "-Inf")
```

```
[1] "-Inf_nil_income"
```

```
str_replace("Neg_nil_income", "nil", "0")
```

```
[1] "Neg_0_income"
```

```
str_replace("we_want_fourthings", "rt", "r_t")
```

```
[1] "we_want_four_things"
```

```
str_remove("we_want_to_remove_the_extra_stuff", "to_remove_the_extra_")
```

```
[1] "we_want_stuff"
```

Exercise 2

Tidy your census data

Start with loading the `tidyverse` package and setting the path ways to the data.

```
library(tidyverse)
census_path <- here::here("data/2021_GCP_all_for_VIC_short-header/2021 Census GCP All Geograph

SA1_paths <- glue::glue(census_path, "{geo}/VIC/2021Census_G17{alpha}_VIC_{geo}.csv",
                        geo = "SA1", alpha = c("A", "B", "C"))

STE_paths <- glue::glue(census_path, "{geo}/VIC/2021Census_G17{alpha}_VIC_{geo}.csv",
                        geo = "STE", alpha = c("A", "B", "C"))
```

To load the data we will use the `here` function from the `here` package. The path argument parsed to `here::here` will be referenced with respect to the `.Rproj` file. This is particularly helpful if your project structure becomes deeply nested.

```
tutorial-04
  analysis
    exercise-04.Rmd      # << analysis are here
  data                  # << data are here

tutorial-04.Rproj # << make reference point `here`
```

First focus on `2021Census_G17A_VIC_STE.csv`, and develop an understanding of what R reads in using the function `str`. This will help you develop an idea of the data structure.

```
STE_G17A <- read_csv(STE_paths[1])
str(STE_G17A)
```

- How many rows and columns does `STE_G17A` contain?

Part of our mission today is to make this data into a tidy form. The first step to this is pivoting the data from it's wide format into a long format. Check out the [tidyverse cheat sheet](#) for some extra help understanding how the data is being combined.

```
STE_G17A_long <- STE_G17A |>
  pivot_longer(cols = -1, names_to = "category",
               values_to = "count")
```

Use `View()` to compare `STE_G17A` before and `STE_G17A_long` after.

- What about the other `STE_paths`? Is the data structured similarly?
The answer is yes, so let's read in all the data and combine it together.

```

data_paths <- STE_paths

# Read in each of the three tables
tbl_G17A_long <- read_csv(data_paths[1]) |>
  pivot_longer(cols = -1, names_to = "category",
               values_to = "count")

tbl_G17B_long <- read_csv(data_paths[2]) |>
  pivot_longer(cols = -1, names_to = "category",
               values_to = "count")

tbl_G17C_long <- read_csv(data_paths[3]) |>
  pivot_longer(cols = -1, names_to = "category",
               values_to = "count")

# Combine all the data together
tbl_G17_long <- bind_rows(tbl_G17A_long, tbl_G17B_long, tbl_G17C_long)

```

We can sanity check our operations by looking at the changes in the dimension of the data.

```
nrow(tbl_G17A_long) + nrow(tbl_G17B_long) + nrow(tbl_G17C_long)
```

```
[1] 510
```

```
nrow(tbl_G17_long)
```

```
[1] 510
```

The next step is to separate the `category` column out into multiple different variables for `sex`, `age_min`, `age_max`, `income_min`, and `income_max`. There is a handy function that can help us do that called `separate_wider_delim()`, which is similar to `str_split()` but works for splitting columns in data frames.

The catch with using `separate_wider_delim()` is that we want all entries in the column to be of a similar format. However, there are a lot of weird cases that we would need to change to make look more standard.

- 1) `Neg_Nil_income` → change to `-Inf_0`.
 1*) `Negtve_Nil_income` → change to `-Inf_0`.
- 2) `more` → `Inf`.
- 3) `PI_NS` → don't want these columns for today (like `NA_NA`).

- 4) 85ov → change to 85_110. 4*) 85_yrs_ov → change to 85_110.
- 5) Tot → don't want these columns (can reproduce them from the data).

One way to find these edge cases is by looking through the meta data. Another is to count the number of underscores of each entry in the column `category`.

```
underscore_count_per_category = str_count(string = tbl_G17_long$category, pattern = "_")
table(underscore_count_per_category)
```

```
underscore_count_per_category
 2   3   4   5   6
6  84  30 339  51
```

- Remove any columns with the header that contains the strings Tot (totals) or PI_NS (personal income not stated).

```
tbl_G17_long_formatted <- tbl_G17_long |>
  filter(!str_detect(string = category, pattern = "Tot"),
         !str_detect(category, "PI_NS"))
```

- Wrangle your data so that it has the columns: `count`, `sex`, `age_min`, `age_max`, `income_min`, and `income_max`. For those with no upper bound (e.g. 85 over, 3000 or more), you can use `Inf` in R to signify ∞ . A quick google tells us no one in Australia is older than 110, so that seems like a more reasonable upper bound for age.

```
tbl_G17_long_formatted <- tbl_G17_long_formatted |>
  mutate(
    category = str_replace(category, "Neg_Nil_income", "-Inf_0"),
    category = str_replace(category, "Neg_Nil_incme", "-Inf_0"),
    category = str_replace(category, "Negtve_Nil_incme", "-Inf_0"),
    category = str_replace(category, "more", "Inf"),
    category = str_replace(category, "85ov", "85_110_yrs"),
    category = str_replace(category, "85_yrs_ovr", "85_110_yrs"))
```

We can do a quick check to see if our changes make the entries in columns `category` have the same number of “_” separators.

```
underscore_count_per_category = str_count(tbl_G17_long_formatted$category, "_")
table(underscore_count_per_category)
```

```
underscore_count_per_category
5
405
```

Looks good, so let's separate out those different entries into columns. Remember to check your result using `View()`.

```
tbl_G17_tidy <- tbl_G17_long_formatted |>
  mutate(category = str_remove(category, "_yrs")) |>
  separate_wider_delim(cols = category, delim = "_",
                      names = c("sex", "income_min", "income_max", "age_min", "age_max"))
# View(tbl_G17_tidy)
```

- What is the importance of the `mutate` call using `str_remove()`?

Later on we are going to do some graphing using different categorical ranges. So let's also create two new variables for age bracket and income bracket.

```
tbl_G17_tidy = tbl_G17_tidy |>
  unite("income", c(income_min, income_max), remove = FALSE) |>
  unite("age", c(age_min, age_max), remove = FALSE)
# View(tbl_G17_tidy)
```

- Repeat the above for the SA1 data. This should be as easy as changing `data_paths` from `STE_paths` to `SA1_paths`.

Take time to note here just how many different steps we needed to do to wrangle our data into a useful format for our filtering and plotting needs. Figuring out all these steps yourself would take a lot of time! To get quicker at understanding what is in your data and how to format it for an analysis will come with time and practice.

Exercise 3

Quality check your census data

- According to the data from Exercise 2, how many people are there in Victoria? If you check <http://www.population.net.au>, there are 6.62 million people in 2021 at Victoria. Does it look right? Are we missing some people in the data? If so, what kinds of people are we missing?
- What is the minimum and maximum of values for `count`? Do the range and categories for each column look right?

In your own time: Exercise 4

- Draw a barplot (using `ggplot` or otherwise) of the number people in Victoria by:
 1. sex
 2. age group
 3. income group
 4. sex and age group
 5. sex and income group
 6. age and income group
 7. sex, age and income group

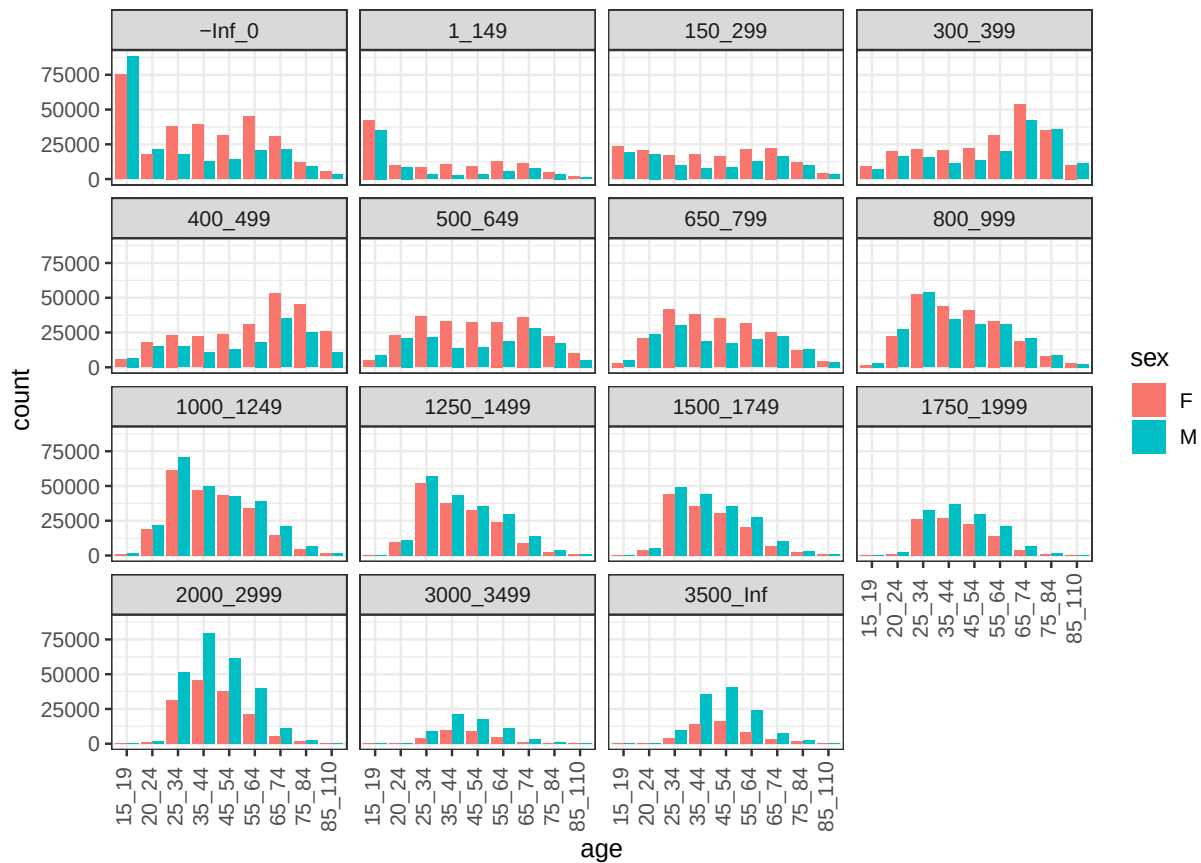
Note to make your labels appear in the correct order on your axes you will need to run the following code.

```
tbl_G17_tidy$income <- fct_reorder(tbl_G17_tidy$income,  
                                   as.numeric(tbl_G17_tidy$income_min))
```

Hint: [click here](#) for a code for the 7th barplot

```
tbl_G17_tidy %>%  
  filter(sex != "P") |>  
  ggplot(aes(x = age, y = count, fill = sex)) +  
  geom_col(position = "dodge") +  
  facet_wrap(~income) +  
  theme_bw(base_size = 12) +  
  theme(axis.text.x = element_text(angle = 90, vjust = 0.3)) +  
  ggtitle("7. Income, sex & age distribution in Victoria from 2021 Census")
```

7. Income, sex & age distribution in Victoria from 2021 Census



- Do the graphs meet your expectations of the data? Discuss with your classmates.
- Is it easier to answer these questions because we made our data tidy?

In your own time: Exercise 5

Extract statistics from your tidy census data

According to the 2021 Census data:

- How many women in Victoria are aged between 15-54 years old?
- What is the proportion of people in Victoria that are 25-34 years old (inclusive) and earn \$1750 or more per week?
- Suppose I randomly select a man from all the men aged 25-44 years old in Victoria. What is the probability that the man I selected earns less than \$1500 per week?

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